NINE PROPERTIES OF MORPHISMS OF ALGEBRAIC SPACES ARE INSENSITIVE TO A THICKENING OF SOURCE AND TARGET

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2000-2016 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 2018-2041, the complete author proof (22 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Morphisms of thickenings of algebraic spaces, etale charts, and the schemes version of the thickening lemma; the reductions are retained.

Source reference. Original lemma lemma-base-change-thickening: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 1942–1949. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-thickening-scheme: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 1730–1733. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma spaces-limits-proposition-affine: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2041–2041. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma spaces-morphisms-lemma-integral-universally-closed: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2041–2041. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $(f, f') : (X \subset X') \to (Y \subset Y')$ be a morphism of thickenings of algebraic spaces over S. Then*

(1) *f is an affine morphism if and only if f' is an affine morphism,*
(2) *f is a surjective morphism if and only if f' is a surjective morphism,*

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

(3) *f is quasi-compact if and only if f' quasi-compact,*
(4) *f is universally closed if and only if f' is universally closed,*
(5) *f is integral if and only if f' is integral,*
(6) *f is (quasi-)separated if and only if f' is (quasi-)separated,*
(7) *f is universally injective if and only if f' is universally injective,*
(8) *f is universally open if and only if f' is universally open,*
(9) *f is representable if and only if f' is representable, and*
(10) *add more here.*

Proof. Observe that $Y \to Y'$ and $X \to X'$ are integral and universal homeomorphisms. This immediately implies parts (2), (3), (4), (7), and (8). Part (1) follows from Limits of Spaces, Proposition spaces-limits-proposition-affine which tells us that there is a 1-to-1 correspondence between affine schemes étale over X and X' and between affine schemes étale over Y and Y'. Part (5) follows from (1) and (4) by Morphisms of Spaces, Lemma spaces-morphisms-lemma-integral-universally-closed. Finally, note that

$$X \times_Y X = X \times_{Y'} X \to X \times_{Y'} X' \to X' \times_{Y'} X'$$

is a thickening (the two arrows are thickenings by Lemma lemma-base-change-thickening). Hence applying (3) and (4) to the morphism $(X \subset X') \to (X \times_Y X \to X' \times_{Y'} X')$ we obtain (6). Finally, part (9) follows from the fact that an algebraic space thickening of a scheme is again a scheme, see Lemma lemma-thickening-scheme. □